INDIVIDUAL REFLECTION

Mini Use Case 02 – Credit Pre-Screening

Calibo AI Academy | Part 2 – KLU Batch | Phase 2 | Supervised Learning Part 1

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Role	Model Builder & Policy Analyst
Contribution	Step 5.4: Logistic Regression Pre-Screener, Step 5.5: Threshold Sweep, Step 5.6: Fairness Check & Step 5.7: Credit Policy Briefing

Business Objective

Use six months of historical loan decisions to identify clear-cut declines that may be removed from manual review, while limiting the risk of wrongly turning away creditworthy applicants. The aim is to balance safe screening with analyst time savings—not simply to maximize model accuracy.

Step 5.4 — Logistic Regression Pre-Screener

1. Target and features. The target was encoded as Declined = 1 and Approved = 0. Features included credit score, new-to-credit flag, FOIR, loan-to-income, delinquencies, active loans, job tenure, age, loan tenure, city tier, employment type and loan purpose. Decision was the target, not a feature; application ID, date, city and gender were excluded from model inputs.

2. Train-test split. An 80/20 stratified split with random_state=42 produced 9,600 training rows and 2,400 test rows. Training data fitted the model and preprocessing; the held-out test data measured performance.

3. Preprocessing. Missing income was filled using employment-type medians with an overall fallback. Missing credit scores were filled with the training median while the is_ntc flag preserved the fact that those applicants had no score. FOIR = (existing EMI + proposed EMI) / monthly income; loan-to-income = loan amount / (monthly income × 12). Numeric features were scaled and employment type and loan purpose were one-hot encoded. A pipeline kept these transformations together with Logistic Regression.

Model results at the default 0.50 threshold:

Metric	Result	Meaning
Accuracy	79.08%	Share of all test predictions that were correct.
ROC-AUC	0.867	Ability to rank declines above approvals.
Precision — Declined	79.11%	Share of predicted declines that were actually declined historically.
Recall — Declined	75.47%	Share of actual declines identified by the model.

Confusion matrix:

Actual	Predicted Approved	Predicted Declined
Approved	1,046 (TN)	225 (FP - Wrongly Declined)
Declined	277 (FN)	852 (TP)

The key risk at the default threshold is the 225 false positives: applicants historically approved by the bank but predicted as declined. This is why accuracy alone is insufficient for a lending decision.

Important coefficients:

Feature	Coefficient	Interpretation
FOIR	+1.22	Higher repayment obligations relative to income push toward decline.
Credit score	−1.01	Higher scores push toward approval.
Delinquencies in 24 months	+0.98	More missed or late repayments push toward decline.
Gig / Contract employment	+0.62	This category pushes toward decline in the fitted model.
Salaried employment	−0.45	This category pushes toward approval in the fitted model.

Step 5.5 — Auto-Decline Threshold

The model produces a probability of decline. A threshold determines how high that probability must be before an application is flagged for automatic decline. The assignment requires that no more than 5% of the auto-declined group were historically approved.

Threshold	Auto-declined	Wrongly declined	Policy Compliance (≤ 5%)
0.50	44.88%	20.89%	Fails (severe risk violation)
0.60	37.62%	16.83%	Fails
0.70	30.75%	12.06%	Fails
0.80	23.46%	7.10%	Fails
0.85	19.67%	5.93%	Borderline / Fails unrounded rule
0.90 (Selected)	**15.21%**	**4.66%**	**Passes (Complies with 5% risk cap)**
0.95	10.00%	2.08%	Passes (Overly conservative)

Monthly business impact:

- Applications screened: 3,000 × 15.21% ≈ 456 applications per month.
- Analyst hours saved: 456 × 25 minutes ÷ 60 ≈ 190 hours per month.
- Observed wrong-decline rate: 4.66% (17 of 365 auto-declined test applications were historically approved).
- **New-to-credit safeguard:** The test results include 12 new-to-credit applicants in the automatic-decline group. Since accuracy for this group is only 56.7%, they must remain with human analysts.

Step 5.6 & 5.7 — Fairness Check and Credit Policy Briefing

I evaluated gender patterns across the dataset. Historical approval was 54.1% for males and 50.4% for females (a 3.7 percentage point gap). Looking into applicant composition, female median income was ₹45,000 compared to ₹55,000 for males, and 21.4% of females were in gig roles compared to 11.6% of males. At the 0.90 threshold, the model auto-declines 15.0% of males and 15.6% of females, showing balanced operational impact.

In the credit policy briefing, I recommended a controlled 90-day pilot at the 0.90 threshold with human review strictly retained for all new-to-credit applicants, monthly fairness monitoring by gender, and model re-validation against actual loan repayment/default outcomes before full operational rollout.

Final Summary

My contribution covered Steps 5.4, 5.5, 5.6, and 5.7. I built and evaluated the Logistic Regression classifier, documented the flaw of the default 0.50 cutoff, selected the 0.90 threshold to satisfy the 5% risk policy, audited gender fairness, and synthesized findings into the final policy recommendations. The pre-screener can safely automate ~456 cases/month and free ~190 analyst hours under supervised human safeguards.